Cognitive Visual Lab · Plate 02

Spaced Repetition: Warming the Memory Cache

The Only Scientific Strategy Against Forgetting · Printable Plate

Ebbinghaus Forgetting Curve · Desirable Difficulty

I. THE CURVE

Fig. 1

Retention Rate Over Time

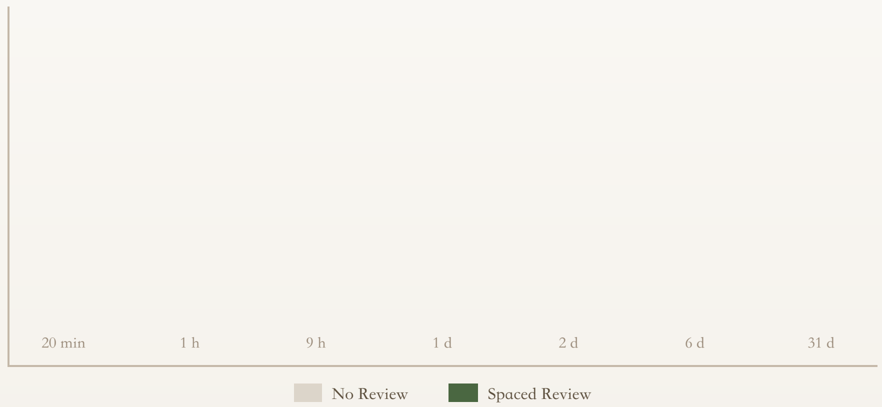
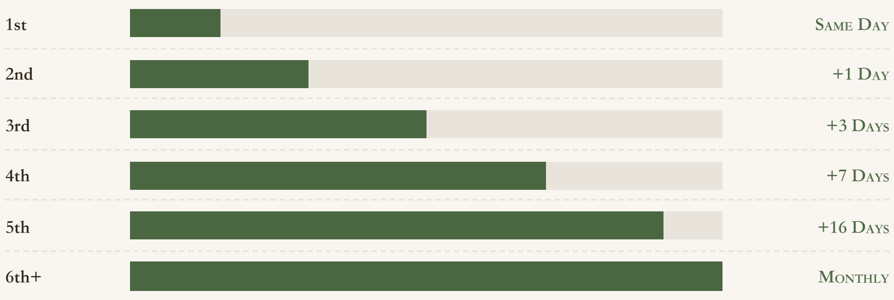


Fig. 2

Optimal Review Schedule



** Core mechanism: desirable difficulty —— reviewing at the edge of recall yields optimal results*

II. OBSERVATIONS

Obs. 1

Forgetting is Front-Loaded

Ebbinghaus's experiment (nonsense syllables) shows massive forgetting within 1 hour, but the rate slows dramatically after 1 day. Meaningful content decays slower, but the pattern holds — the first review is the most critical.

Obs. 2

Spacing > Massing

The same study time spread over 10 days is 2-3× more effective than crammed into 1 day. Continuous reading creates an "illusion of fluency."

Obs. 3

Retrieval > Rereading

Active recall (self-testing with the book closed) is significantly more effective than passive rereading. Testing is a learning tool, not just assessment — it strengthens neural connections in "hard mode."

Obs. 4

Review at the Edge of Forgetting

Reviewing when you're "about to forget but haven't yet" yields the strongest memory consolidation. Too frequent wastes effort; too sparse loses progress.

METHODS

1. **Review before sleep** — Active recall 8-12 hours after learning more than doubles retention. During sleep, the brain actively reorganizes and consolidates memories.
2. **Use spaced repetition software** — Anki, RemNote, etc. have built-in algorithms that schedule optimal review intervals automatically.
3. **Test yourself first, then check** — Forcing your brain to retrieve in "hard mode," even if wrong, is more effective than immediately seeing the answer.
4. **Interleave different topics** — Don't study one chapter at a time; mix reviews. The brain builds stronger connections through switching.
5. **Stop new learning 1 hour before sleep** — During sleep, the hippocampus "writes" cached memories to cortical long-term storage. New information interferes with this consolidation.